

Satoru Akita

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SUMMARY

- Software Engineer specializing in **edge AI firmware, embedded systems, and IT infrastructure** — full stack from sensor protocol to cloud.
- Currently leading embedded FW development for factory AI cameras and driving generative AI adoption — from tool selection to engineering culture — for a new org.
- 7+ years at Sony Semiconductor: edge AI, computer vision, enterprise AI deployments, and OSS leadership across international partnerships.

SKILLS

Languages: Python, C/C++, TypeScript, JavaScript, Bash

AI / ML: PyTorch, TensorFlow, OpenCV, Computer Vision, Anomaly Detection, Edge AI

Cloud & Infra: Azure, AWS, GCP, Docker, CI/CD, GitHub Actions, MLOps

Hardware: Raspberry Pi, IMX500, ESP32, switching hub config, MEMS sensors

Soft Skills: Solution scoping, on-site technical diagnosis, HW–SW integration, SW event operations

EXPERIENCE

Senior Software Engineer | Sony Semiconductor Solutions, Tokyo

Apr 2026 – Present

- Developing factory AI camera FW with **GenICam**-based frame acquisition, C++ anomaly-detection inference, and **SLMP** result transmission; includes full-stack operator web app
- Led evaluation and adoption of **generative AI tools, CI/CD pipelines, remote access infrastructure, and GitHub repository strategy** for a newly formed team

AI Solutions Engineer / Architect | Sony Semiconductor Solutions, Tokyo

Jan 2023 – Mar 2026

Edge AI Sample Applications

- Led **6-engineer team** to develop and release **9 OSS Edge AI sample apps**; featured in exhibitions, national project demos, and enterprise briefings across Japan
- Drove international OSS collaboration (**Raspberry AI Camera**) with Lund (Sweden), Raspberry Pi Foundation, and Midokura (Spain)

Edge–Cloud AI Deployment

- Deployed multi-camera edge–cloud AI systems across **12 customer environments** (10+ cameras/site), coordinating **5 engineers + 5 cross-functional members**
- Delivered AI solutions for **8+ enterprise clients**: license plate recognition, shelf vacancy detection, traffic heatmaps, digital signage analytics

Generative AI & Engineering Productivity

- Led **5-engineer team** to build and OSS-release AITRIOS MCP server; automated **~80% of API migration work** (V1 → V2) via LLM-assisted scripts

Edge AI Embedded Engineer | Sony Semiconductor Solutions, Tokyo

Jan 2020 – Dec 2022

- Developed SDK and first-gen Edge AI demos for Sony's **IMX500**; served as technical lead for **2 exhibition deployments** (system setup, live operation, on-site client briefings)
- Designed full-stack on-premise camera systems covering circuit review, SoC research, and HW validation

Computer Vision / ML Engineer | Sony Semiconductor Solutions, Tokyo

Apr 2019 – Dec 2019

- Independently built **46-class sign language recognition (98% accuracy, 15 FPS)** using ToF + facial landmark fusion — data collection to deployment
- Led ML dev of ToF+RGB 3D pose estimation; **exhibited at CES 2019**

EDUCATION

M.S. in Robotics

Tohoku University 2017–2019

MEMS sensors, semiconductor processes, data utilization, machine learning

B.Eng. in Mechanical Engineering

Tohoku University 2013–2017

Molecular robotics (DNA nanotechnology); published in *Small* (2017); robot contests, particle simulation (C++ / OpenGL)

CERTIFICATIONS

Google Certified Professional – Cloud Architect

TensorFlow Developer Certificate

AWS Solutions Architect – Associate

Azure AI Engineer Associate

Registered Information Security Specialist (Japan)

AWARDS & PUBLICATIONS

SecHack365 (2019)

NICT-sponsored 365-day security hackathon. Built Fasttext AI to visualize emotional transitions in Twitter data. Selected for SXSW (Austin) — won corporate prize at on-site hackathon.

Grand Prize, BIOMOD 2015

International biomolecular design competition, Harvard University

3rd Place Worldwide, iCAN'14

International embedded systems competition

Publications:

Froemel, Akita et al. *Micromachines* (2020)

Uchida, Akita et al. *Small* (2017)

LEADERSHIP & COMMUNITY

SW Event Operations — SRE NEXT, Open Source Summit Japan, SecHack365

Gemini API Hackathon Mentor — Google AI Student Ambassadors

Founding Member — Whole Brain Architecture Young Researchers (Tohoku Chapter, 2018)

Hackathon Enthusiast — Participant across domains incl. humanoid robotics, edge AI, mobile apps, NASA Space Apps Challenge, OpenCV AI competition